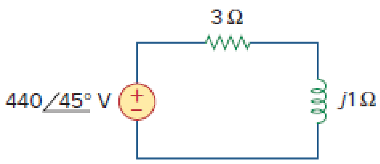
Chapter 11, Problem 3PP

Problem

In the circuit of Fig. 11.4, calculate the average power absorbed by the resistor and inductor. Find the average power supplied by the voltage source.

Figure 11.4:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.4 in the text book.

Calculate the current I flowing in the circuit.

$$\mathrm{I} = \frac{\mathrm{V}}{R + jX_L}$$

Substitute $440\angle 45°\ \mathrm{V}$ for V, $3\ \Omega$ for R and $j1\ \Omega$ for jX_L.

$$\begin{aligned}\mathrm{I} &= \frac{440\angle 45°}{3 + j1} \\ &= \frac{440\angle 45°}{3.162\angle 18.44°} \\ &= 139.15\angle 26.56°\ \mathrm{A}\end{aligned}$$

Therefore, the value of the current I flowing in the circuit, is $139.15\angle 26.56°\ \mathrm{A}$.

Step 2 of 3

Calculate the average power P_R absorbed by the resistor.

$$P_R = \frac{1}{2}|\mathrm{I}|^2 R$$

Substitute $139.15\ \mathrm{A}$ for $|\mathrm{I}|$ and $3\ \Omega$ for R.

$$\begin{aligned}P_R &= \frac{1}{2}(139.15)^2(3) \\ &= (19.36\ \mathrm{k})(1.5) \\ &= 29.04\ \mathrm{kW}\end{aligned}$$

Therefore, the value of the power P_R absorbed by the resistor is, $\boxed{29.04\ \mathrm{kW}}$.

The value of the average power absorbed by the inductor is, $\boxed{0\ \mathrm{W}}$.

Step 3 of 3

Calculate the value of the average power P delivered by the voltage source.

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Substitute 139.15 A for I_m, 440 V for V_m, $45°$ for θ_v and $26.56°$ for θ_i.

$$\begin{aligned} P &= \frac{1}{2}(440)(139.15)\cos(45° - 26.56°) \\ &= (30.61 \text{ k})\cos(18.44°) \\ &= (30.61 \text{ k})(0.9487) \\ &= 29.04 \text{ kW} \end{aligned}$$

Therefore, the value of the average power P delivered by the voltage source is, $\boxed{29.04 \text{ kW}}$

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